Venous Thromboembolism: DVT and PE

All key points for the exam (CHEST, ASH, ESC) — drug table at the end

1. Pathophysiology and Risk Factors

- **VTE** = **deep vein thrombosis (DVT)** + **pulmonary embolism (PE)** • most PEs come from **proximal leg DVT** (popliteal, femoral, iliac veins).
- **Virchow's triad: venous stasis** (immobility, paralysis, long travel, HF), **vascular (endothelial) injury** (surgery, trauma, catheters, prior DVT), **hypercoagulability** (cancer, estrogen, pregnancy, thrombophilia).
- **Coagulation cascade targets:** UFH (antithrombin → IIa = Xa) • LMWH (Xa > IIa) • fondaparinux (indirect, Xa only) • rivaroxaban, apixaban, edoxaban (direct Xa) • dabigatran, argatroban, bivalirudin (direct thrombin IIa) • warfarin (vitamin K–dependent factors **II, VII, IX, X** and proteins C and S).

Category	Risk factors
Major transient (provoked)	**Major surgery** (especially orthopedic hip/knee, cancer surgery) with general anesthesia > 30 min, **trauma**, fractures, hospitalization with acute illness and bed rest ≥ 3 days, C-section
Minor transient	**Estrogen therapy** (combined oral contraceptives, HRT), **pregnancy/postpartum**, long-distance travel, leg injury with reduced mobility, minor surgery, hospitalization < 3 days
Persistent	**Active cancer** (pancreas, brain, stomach, lung, ovary, hematologic), **prior VTE**, **thrombophilia**, inflammatory disease (IBD, SLE), obesity, age > 60, HF, nephrotic syndrome, myeloproliferative disorders, paralysis
Inherited thrombophilia	**Factor V Leiden** (most common, activated protein C resistance), **prothrombin G20210A**, deficiencies of **protein C**, **protein S**, **antithrombin** (heparin resistance)
Acquired thrombophilia	**Antiphospholipid syndrome** (lupus anticoagulant, anticardiolipin, anti-β2-glycoprotein I) • HIT • PNH
Drugs	**Estrogens** (COCs — highest with 3rd-gen progestins/drospirenone, patch, ring), **SERMs** (**tamoxifen**, **raloxifene**), testosterone, **ESAs** (epoetin), **thalidomide/lenalidomide** (with dexamethasone), **L-asparaginase**, **bevacizumab**, **JAK inhibitors** (tofacitinib — boxed), antipsychotics, glucocorticoids, tranexamic acid, heparin (HIT)

- **Unprovoked** VTE = no identifiable risk factor → **higher recurrence risk** (~10% in year 1, ~30% at 5 years) → consider extended therapy.
- Thrombophilia testing: not routinely needed; avoid during the acute event and on anticoagulants (protein C/S, antithrombin falsely low; DOACs interfere with lupus anticoagulant tests).

2. Deep Vein Thrombosis (DVT)

Clinical presentation

- **Unilateral leg swelling**, **pain/tenderness** along the deep veins, warmth, erythema, dilated superficial veins, calf circumference difference > 3 cm • Homans' sign unreliable • many are asymptomatic.
- **Proximal DVT** (popliteal and above) → high PE risk → always treat • **isolated distal (calf) DVT** → lower risk.
- **Phlegmasia cerulea dolens** (massive iliofemoral DVT → limb ischemia, cyanosis) → emergency (catheter-directed thrombolysis/thrombectomy).
- **Complications:** PE, **post-thrombotic syndrome** (chronic pain, swelling, skin changes, ulcers in 20–50%), recurrence.

Diagnosis — Wells score for DVT

Criterion	Points
Active cancer (treatment within 6 months or palliative)	+1
Paralysis, paresis, or recent plaster cast of the leg	+1
Recently bedridden ≥ 3 days or major surgery within 12 weeks	+1
Localized tenderness along the deep venous system	+1
Entire leg swollen	+1
Calf swelling > 3 cm compared with the other leg	+1
Pitting edema confined to the symptomatic leg	+1
Collateral (non-varicose) superficial veins	+1
Previously documented DVT	+1
Alternative diagnosis at least as likely as DVT	**−2**

- **Wells ≤ 1 = DVT unlikely** → **D-dimer** → if **negative, DVT excluded**; if positive → compression ultrasound.
- **Wells ≥ 2 = DVT likely** → **compression ultrasound (CUS)** directly (± D-dimer to decide on repeat ultrasound if negative).
- **D-dimer** (fibrin degradation product): **high sensitivity, low specificity** → useful to **rule out** in low-probability patients only • ↑ in cancer, pregnancy, infection, surgery, elderly, inflammation • **age-adjusted cut-off = age × 10 µg/L** for patients > 50.
- **Compression ultrasound** = imaging test of choice for DVT (non-compressible vein).

3. Pulmonary Embolism (PE)

Clinical presentation

- **Dyspnea** (most common, sudden), **pleuritic chest pain**, **tachypnea**, **tachycardia**, **hypoxemia**, cough, **hemoptysis**, syncope, signs of DVT.
- Massive PE: **hypotension/shock**, **RV failure** (JVD), cardiac arrest (PEA).
- **ABG:** hypoxemia, **respiratory alkalosis** (hypocapnia), ↑ A–a gradient.
- **ECG: sinus tachycardia** (most common), **S1Q3T3** (classic but uncommon), right bundle branch block, T-wave inversions V1–V4 (RV strain), new AF.
- **Chest X-ray:** usually normal or nonspecific (Westermark sign, Hampton's hump) — helps exclude other causes.
- **Biomarkers: troponin** (myocardial injury) and **BNP/NT-proBNP** (RV strain) → used for **risk stratification**, not diagnosis.
- **Echocardiography: RV dilation/dysfunction** (McConnell's sign) — key in unstable patients who cannot go to CT.

Diagnosis — Wells score for PE

Criterion	Points
Clinical signs and symptoms of DVT	+3
PE is the most likely diagnosis (alternative less likely)	+3
Heart rate > 100 bpm	+1.5
Immobilization ≥ 3 days or surgery in previous 4 weeks	+1.5
Previous DVT or PE	+1.5
Hemoptysis	+1
Malignancy (treatment within 6 months or palliative)	+1

- **Wells ≤ 4 = PE unlikely** → **D-dimer** (age-adjusted) → negative = PE excluded; positive → **CT pulmonary angiography (CTPA)**.

- **Wells > 4 = PE likely** → **CTPA** directly (D-dimer not useful) • start anticoagulation while waiting if high probability and low bleeding risk.
- **PERC rule** (in **low clinical probability** only — if **all 8 negative**, no further testing): age **< 50**, HR **< 100**, SaO2 **≥ 95%**, no hemoptysis, no estrogen use, no prior VTE, no unilateral leg swelling, no surgery/trauma requiring hospitalization within 4 weeks.
- Other algorithms: **YEARS** (3 items: DVT signs, hemoptysis, PE most likely + D-dimer threshold 500 or 1,000) • revised Geneva score.
- **CTPA** = test of choice • **V/Q scan** when contrast is contraindicated (**severe CKD**, **contrast allergy**, **pregnancy** with normal CXR) • **leg compression ultrasound** showing DVT + PE symptoms → treat without CTPA.
- **Unstable (shock)**: **bedside echo** showing RV dysfunction is enough to justify reperfusion if CTPA not possible.
- **Pregnancy:** CUS of the legs first if DVT symptoms → CXR → V/Q (normal CXR) or CTPA • YEARS-adapted algorithm.

Risk stratification (ESC)

Risk class	Features	Treatment
High risk ("massive")	**Hemodynamic instability**: cardiac arrest, **obstructive shock** (SBP < 90 mmHg or vasopressors with end-organ hypoperfusion), or **persistent hypotension** (SBP < 90 or drop ≥ 40 mmHg for > 15 min)	**UFH + systemic thrombolysis** (alteplase) • if contraindicated/failed → **surgical embolectomy or catheter-directed therapy** • ECMO • norepinephrine/dobutamine, cautious fluids (≤ 500 mL)
Intermediate-high ("submassive")	Normotensive, PESI III–V or sPESI ≥ 1 **+ RV dysfunction** (echo/CT) **+ ↑ troponin**	**Anticoagulation** (LMWH/UFH initially) + **close monitoring** → **rescue thrombolysis** if hemodynamic decompensation • routine full-dose thrombolysis **not** recommended (PEITHO: ↑ bleeding/ICH) • catheter-directed therapy in selected
Intermediate-low	sPESI ≥ 1 with **only one or none** of RV dysfunction / ↑ troponin	**Anticoagulation** (DOAC or LMWH), hospital
Low risk	**sPESI 0** (or PESI I–II), no RV dysfunction, normal troponin	**Anticoagulation (DOAC) • early discharge/outpatient** treatment (Hestia criteria) if good social support

- **sPESI** (1 point each): **age > 80**, **cancer**, **chronic cardiopulmonary disease** (HF or chronic lung disease), **HR ≥ 110**, **SBP < 100**, **SaO2 < 90%** → **0 = low risk** (30-day mortality ~1%), ≥ 1 = higher risk.

4. Anticoagulant Treatment of VTE — Overview

Treatment phases

Phase	Duration	Purpose
Initiation	First 5–21 days	Higher-intensity therapy (parenteral lead-in, or apixaban 10 mg BID × 7 days / rivaroxaban 15 mg BID × 21 days)
Primary (treatment)	**3 months** (minimum for all VTE)	Prevent extension, embolization, early recurrence
Extended	Beyond 3 months, no planned stop date (reassess yearly)	Prevent late recurrence (unprovoked, persistent risk factors) — reduced-dose DOAC possible

Choice of anticoagulant (CHEST 2021)

- **DOACs (apixaban, rivaroxaban, edoxaban, dabigatran) are preferred over warfarin** for most patients (non-inferior efficacy, **less major bleeding/ICH**, no INR monitoring).
- **Apixaban and rivaroxaban = oral-only** regimens (no parenteral lead-in) • **dabigatran and edoxaban require 5–10 days of parenteral** anticoagulation first.
- **Warfarin** requires **parenteral overlap ≥ 5 days AND INR ≥ 2 for ≥ 24 h** (two consecutive INRs).

- **Start anticoagulation immediately** when VTE is confirmed (or suspected with high probability and low bleeding risk while awaiting tests).
- **Home treatment** preferred for DVT and low-risk PE (if stable, low bleeding risk, adequate support/access).
- **Early ambulation** recommended (not bed rest).

Situation	Preferred anticoagulant
Most patients (no cancer)	**DOAC** (apixaban, rivaroxaban, edoxaban, dabigatran) over warfarin
Cancer-associated VTE	**Oral Xa inhibitor (apixaban, rivaroxaban, edoxaban)** preferred over LMWH • **LMWH** (dalteparin, enoxaparin) if **luminal GI/GU cancer** (↑ bleeding with edoxaban/rivaroxaban — apixaban acceptable), significant **drug interactions** (chemotherapy), poor GI absorption, thrombocytopenia • treat ≥ 6 months and as long as cancer active
Antiphospholipid syndrome	**Warfarin (INR 2–3)** — **avoid DOACs** (especially **triple-positive**: ↑ arterial thrombosis)
Pregnancy / breastfeeding	**LMWH** (does not cross placenta) • **warfarin teratogenic** (CI in pregnancy; OK in breastfeeding) • **DOACs avoided** in both
Severe renal impairment (CrCl < 30) / dialysis	**UFH → warfarin** • apixaban (labeled without adjustment, used in ESRD) • avoid dabigatran, edoxaban, rivaroxaban (< 15–30), fondaparinux (< 30) • enoxaparin 1 mg/kg q24h if CrCl < 30
Hemodynamically unstable PE / possible thrombolysis or procedure	**IV UFH** (short half-life, fully reversible with protamine)
Severe hepatic impairment (Child-Pugh B/C)	LMWH (avoid DOACs in Child-Pugh C; rivaroxaban/edoxaban avoid in B)
Obesity (BMI > 40 or weight > 120 kg)	**Apixaban or rivaroxaban** at standard doses (ISTH 2021) • or LMWH weight-based (anti-Xa monitoring) / warfarin • avoid dabigatran/edoxaban
After bariatric surgery	Parenteral or warfarin initially (DOAC absorption uncertain)
Heparin-induced thrombocytopenia	**Non-heparin**: argatroban, bivalirudin, fondaparinux, DOACs
Poor adherence / cost concerns / mechanical valve	Warfarin (INR monitoring allows adherence check)

Duration of anticoagulation

VTE type	Recommended duration
Provoked by major transient risk factor (surgery)	**3 months** then stop
Provoked by minor transient risk factor (estrogen, travel)	**3 months** (stop the risk factor, e.g., estrogen)
Unprovoked VTE (first episode)	3 months, then **extended (indefinite)** if **low–moderate bleeding risk** • stop at 3 months if high bleeding risk • consider **reduced dose** after 6 months: **apixaban 2.5 mg BID** or **rivaroxaban 10 mg daily**
Persistent risk factor / active cancer	**Extended** (as long as cancer active or risk persists)
Second unprovoked VTE	**Indefinite** (extended) • stop only if high bleeding risk
Isolated distal (calf) DVT	Without severe symptoms/risk factors: **serial ultrasound for 2 weeks** (anticoagulate if extends) • with severe symptoms or risk factors (cancer, prior VTE, inpatient): anticoagulate **3 months**
Subsegmental PE without proximal DVT, low recurrence risk	**Clinical surveillance** (no anticoagulation) acceptable • anticoagulate if high risk
Upper-extremity DVT	**3 months** • catheter-related: keep catheter if functional and needed; anticoagulate while catheter in place (≥ 3 months)
Superficial vein thrombosis (≥ 5 cm, not near junction)	**Fondaparinux 2.5 mg SC daily × 45 days** (or rivaroxaban 10 mg) • within 3 cm of saphenofemoral junction → treat as DVT
Pregnancy-associated VTE	LMWH for rest of pregnancy and **≥ 6 weeks postpartum** (minimum total 3 months)

- If anticoagulation is stopped after unprovoked VTE, **aspirin 81–100 mg** reduces recurrence modestly (less effective than anticoagulants).
- Reassess extended therapy **at least yearly** (bleeding risk, preferences, renal function).

5. Parenteral Anticoagulants

Unfractionated heparin (UFH)

- **MOA:** binds **antithrombin** → inactivates **thrombin (IIa) and Xa** equally • IV onset immediate • half-life ~60–90 min • **not renally cleared**.
- **VTE treatment: 80 units/kg IV bolus → 18 units/kg/h** infusion (actual body weight) • or SC 333 units/kg then 250 units/kg q12h (fixed, unmonitored).
- **ACS dosing** is lower (60 units/kg bolus max 4,000; 12 units/kg/h max 1,000/h).
- **Monitoring: aPTT 1.5–2.5 × control** (or **anti-Xa 0.3–0.7 units/mL**) 6 h after start/changes • **CBC/platelets** (baseline, then every 2–3 days for HIT).
- **Preferred when: hemodynamically unstable PE**, possible **thrombolysis**/surgery/procedure, **severe renal failure**, high bleeding risk (rapid reversal), obesity extremes (with monitoring).
- **ADRs: bleeding**, **HIT**, **osteoporosis** (long term), **hyperkalemia** (aldosterone suppression), ↑ LFTs.
- **Heparin resistance** (antithrombin deficiency, ↑ factor VIII) → use anti-Xa monitoring.
- **Prophylaxis: 5,000 units SC q8–12h** (q8h for higher risk; 7,500 q8h in obesity).
- **Reversal: protamine 1 mg per 100 units** of UFH given in the previous **2–2.5 h** (max 50 mg; slow IV — hypotension, anaphylaxis — fish allergy, NPH insulin, vasectomy).

Low-molecular-weight heparins (LMWH)

- **MOA:** antithrombin-mediated, **Xa > IIa** (enoxaparin 3–4:1) • predictable SC absorption → **no routine monitoring** • **renally cleared** • less HIT and osteoporosis than UFH.
- **Enoxaparin treatment: 1 mg/kg SC q12h** (preferred in cancer, obesity, PE) or **1.5 mg/kg SC q24h** • **CrCl < 30: 1 mg/kg q24h** • dose on **actual body weight**.
- **Enoxaparin prophylaxis: 40 mg SC daily** (or 30 mg q12h after orthopedic/trauma surgery) • **CrCl < 30: 30 mg daily** • obesity (BMI ≥ 40): 40 mg q12h.
- **Dalteparin:** treatment **200 units/kg SC daily** • **cancer: 200 units/kg daily × 1 month, then 150 units/kg daily** • prophylaxis 5,000 units daily.
- **Anti-Xa monitoring** (peak 4 h after 3rd–4th dose; goal **0.6–1.0** units/mL for q12h, 1–2 for daily) in: **pregnancy**, **severe obesity**, **renal impairment**, low body weight, children.
- **ADRs:** bleeding, **injection-site bruising/hematoma**, thrombocytopenia/HIT (less), hyperkalemia • **boxed: spinal/epidural hematoma** with neuraxial anesthesia/puncture.
- **Administration:** SC in abdomen (2 inches from navel), **do not expel the air bubble** in prefilled syringe, **do not rub** site, alternate sides.
- **Reversal:** protamine — **1 mg per 1 mg enoxaparin** (if < 8 h; 0.5 mg per 1 mg if 8–12 h) — only **~60% neutralization** of anti-Xa activity.
- **Neuraxial timing:** prophylactic enoxaparin — wait **12 h** before catheter placement/removal • treatment dose — **24 h** • restart ≥ 4 h after catheter removal.

Fondaparinux

- **Synthetic pentasaccharide** → antithrombin-mediated **Xa inhibition only** • once daily SC • half-life 17–21 h • renal elimination.
- **Treatment (weight-based): < 50 kg: 5 mg • 50–100 kg: 7.5 mg • > 100 kg: 10 mg** SC daily • **prophylaxis 2.5 mg daily** (CI if weight < 50 kg for prophylaxis).
- **Contraindicated CrCl < 30 • no cross-reactivity with HIT antibodies** (used in HIT, off-label) • **no reversal agent** • boxed spinal/epidural hematoma.
- **Superficial vein thrombosis:** 2.5 mg daily × 45 days.

Direct thrombin inhibitors (parenteral)

- **Argatroban: hepatic** elimination → good in **renal failure** • HIT: **2 mcg/kg/min** (0.5–1.2 mcg/kg/min in hepatic impairment, HF, critically ill, post-cardiac surgery) • aPTT **1.5–3 × baseline** (max 100 s) • **falsely ↑ INR** → when transitioning to warfarin, stop argatroban when INR > 4 on combined therapy, recheck INR in 4–6 h.
- **Bivalirudin: renal** (partially enzymatic) • **PCI** (with or without HIT) • HIT off-label 0.15 mg/kg/h • short half-life (25 min).
- No reversal agents for either.

6. Direct Oral Anticoagulants (DOACs) for VTE

Drug	VTE treatment dose	Extended / reduced dose	Renal and key points
Apixaban (Eliquis) — Xa	**10 mg BID × 7 days → 5 mg BID** (no parenteral lead-in)	**2.5 mg BID after ≥ 6 months**	No VTE renal adjustment (limited data CrCl < 25; used in ESRD) • ~27% renal • with/without food • **least GI bleeding** • can be crushed (NG tube)
Rivaroxaban (Xarelto) — Xa	**15 mg BID × 21 days → 20 mg daily with food** (↑ absorption of 15/20 mg)	**10 mg daily after ≥ 6 months**	**Avoid CrCl < 15** (VTE) • once daily • ↑ GI bleeding • can be crushed with food
Edoxaban (Savaysa) — Xa	**5–10 days parenteral → 60 mg daily • 30 mg** if CrCl 15–50, weight ≤ 60 kg, or certain P-gp inhibitors	—	Avoid CrCl < 15 • AF: **reduced efficacy if CrCl > 95** (boxed) • ↑ GI bleeding
Dabigatran (Pradaxa) — IIa	**5–10 days parenteral → 150 mg BID**	150 mg BID	**Avoid CrCl < 30** (VTE) • **80% renal** • **dyspepsia/GI upset** • **don't open/crush capsules** (↑ absorption 75%) • keep in **original bottle** (use within 4 months) • prodrug (esterase)

DOAC key points

- **Advantages:** fixed dosing, no routine monitoring, rapid onset (2–4 h) and offset, fewer food/drug interactions, less ICH than warfarin.
- **Disadvantages:** short half-life → **missed doses = loss of protection**, cost, renal dosing, limited data in extremes, APS (avoid), mechanical valves (**contraindicated** — dabigatran RE-ALIGN), moderate–severe mitral stenosis.
- **Missed dose:** apixaban/dabigatran (BID) — take within 6 h, otherwise skip • rivaroxaban 15 mg BID (first 21 days) — take immediately even if 2 tablets at once (30 mg/day) • once-daily — take same day, never double next day.
- **Monitoring:** CBC, SCr/CrCl, LFTs at baseline and at least yearly (more often if CrCl < 60 or elderly: every **CrCl/10 months**).
- **Periprocedural:** no bridging needed • hold 1–2 days before low-bleed-risk procedures (CrCl ≥ 30) and 2–4 days for high-risk (longer for dabigatran with renal impairment).

DOAC drug interactions: all are **P-gp substrates**; apixaban and rivaroxaban are also **CYP3A4 substrates**. **Avoid combined strong CYP3A4 + P-gp inhibitors** (ketoconazole, itraconazole, posaconazole, **ritonavir/cobicistat**) with rivaroxaban; apixaban → **↓ dose by 50%** (if on 2.5 mg BID → avoid) • **avoid strong inducers** (**rifampin**, **carbamazepine**, **phenytoin**, **St John's wort**) with all DOACs • dabigatran/edoxaban: P-gp inhibitors (dronedarone, ketoconazole, verapamil, amiodarone, quinidine) ↑ levels • **additive bleeding**: **NSAIDs**, **aspirin/antiplatelets**, **SSRIs/SNRIs**, fish oil, ginkgo — use PPI if combined antiplatelet.

7. Warfarin (Vitamin K Antagonist)

- **MOA:** inhibits **vitamin K epoxide reductase (VKORC1)** → ↓ synthesis of factors **II, VII, IX, X** and **proteins C and S** • full effect takes **5–7 days** (factor II half-life ~60 h) • early INR rise reflects factor VII (half-life 6 h).
- **Initial transient hypercoagulability** (protein C falls first) → **overlap with parenteral anticoagulant ≥ 5 days and INR ≥ 2 for 24 h**.
- **Starting dose: 5 mg daily** (most) • **≤ 5 mg (2.5 mg)** in elderly, malnourished, liver disease, HF, interacting drugs, high bleeding risk, Asian ancestry • **10 mg × 2 days** in healthy outpatients (CHEST option).
- **Goal INR 2.0–3.0** (target 2.5) for VTE • monitor INR every 2–3 days initially, then weekly → up to every 12 weeks when stable • adjust weekly dose by 5–20%.
- **Pharmacogenomics: CYP2C9** (*2, *3 → ↓ metabolism of S-warfarin → lower dose) and **VKORC1** variants (↑ sensitivity) • CYP4F2.
- **ADRs: bleeding** (most important), **skin necrosis** (days 3–8; protein C deficiency, large loading doses), **purple toe syndrome** (cholesterol emboli), **teratogenic** (embryopathy — **contraindicated in pregnancy**; safe in breastfeeding), calciphylaxis, alopecia.
- **Diet:** keep **vitamin K intake consistent** (leafy greens: spinach, kale, broccoli; green tea, enteral feeds) — don't avoid, just be consistent • alcohol binge ↑ INR; chronic use ↓ INR.
- Things that ↑ INR: acute illness, fever, hyperthyroidism, ↓ intake, diarrhea, liver disease, HF exacerbation • ↓ INR: hypothyroidism, ↑ vitamin K intake, missed doses.

Warfarin drug interactions (S-warfarin = CYP2C9; R-warfarin = CYP3A4, 1A2): ↑ INR (inhibitors): amiodarone (↓ warfarin dose 30–50%), **fluconazole**/azoles, **metronidazole**, **sulfamethoxazole-trimethoprim**, fluoroquinolones, macrolides, **fluvoxamine**, fenofibrate/gemfibrozil, statins (minor), acetaminophen (> 2 g/day), levothyroxine (↑ catabolism of factors), omeprazole, cimetidine, isoniazid, tamoxifen, capecitabine, fish oil • **↓ INR (inducers): rifampin**, **carbamazepine**, **phenytoin** (initially ↑ then ↓), **phenobarbital**, **St John's wort**, nafcillin, dicloxacillin, smoking (1A2), **vitamin K**, cholestyramine (↓ absorption), oral nutritional supplements • **↑ bleeding without INR change: NSAIDs**, **aspirin**, **clopidogrel**, **SSRIs**, ginkgo, garlic.

Managing elevated INR and bleeding (CHEST)

Situation	Management
INR above goal but < 4.5, no bleeding	Hold a dose or ↓ dose; recheck
INR 4.5–10, no bleeding	**Hold warfarin**, resume at lower dose when INR therapeutic • **routine vitamin K not recommended**
INR > 10, no bleeding	Hold warfarin + **oral vitamin K 2.5–5 mg**
Major/life-threatening bleeding (any INR)	Hold warfarin + **4-factor PCC (Kcentra) 25–50 units/kg** (by INR) + **vitamin K 5–10 mg IV slow infusion** (over ≥ 20 min — anaphylaxis risk) • FFP if PCC unavailable

DOAC and heparin reversal

- **Dabigatran: idarucizumab (Praxbind) 5 g IV** (2 × 2.5 g) — monoclonal antibody fragment • also dialyzable.
- **Apixaban, rivaroxaban (and edoxaban off-label): andexanet alfa** (recombinant decoy Xa; thrombotic events — boxed; availability limited) **or 4F-PCC 50 units/kg** (commonly used, off-label).
- **Activated charcoal** if DOAC taken within 2–4 h • **UFH/LMWH → protamine • fondaparinux, argatroban, bivalirudin → no specific reversal** (supportive, rFVIIa, dialysis for bivalirudin).

8. Thrombolysis, Catheter Therapy and IVC Filters

Systemic thrombolysis for PE

- **Indication: high-risk (massive) PE** with hypotension/shock and no high bleeding risk • **rescue** in intermediate-high risk PE that **deteriorates** • **not** for most hemodynamically stable patients • **cardiac arrest from suspected PE**.
- **Alteplase 100 mg IV over 2 hours** (peripheral vein preferred) • cardiac arrest: **50 mg IV bolus** (repeat 50 mg in 15 min; continue CPR 60–90 min) • tenecteplase weight-based bolus (off-label for PE in US) • reduced-dose alteplase (50 mg) studied.
- **Heparin with alteplase: hold UFH during the 2-h infusion** → check aPTT at end → **restart UFH without bolus when aPTT < 2 × ULN** (~< 80 s).
- Most benefit within **48 h** of symptom onset (up to 6–14 days).
- Major ADR: **bleeding**, especially **intracranial hemorrhage** (~2%) • orolingual angioedema (ACE inhibitors).

Absolute contraindications	Relative contraindications
• **Any prior intracranial hemorrhage** • Known structural cerebral vascular lesion (AVM) or malignant intracranial neoplasm • **Ischemic stroke within 3 months** • Suspected **aortic dissection** • **Active bleeding** (excluding menses) or bleeding diathesis • Significant closed-head or facial trauma within 3 months • Recent brain or spinal surgery (within 2 months)	• Age > 75 • **Uncontrolled hypertension** (SBP > 180 or DBP > 110) • Ischemic stroke > 3 months ago, dementia • **Major surgery within 3 weeks**, traumatic or prolonged CPR (> 10 min) • Recent internal bleeding (2–4 weeks) • Noncompressible vascular punctures • **Pregnancy** and first postpartum week • **Current oral anticoagulant use** • Active peptic ulcer, advanced liver disease, infective endocarditis

Other interventions

- **Catheter-directed therapy** (local low-dose thrombolysis or mechanical thrombectomy): high-risk PE when systemic thrombolysis is contraindicated or failed; selected intermediate-high risk patients; **iliofemoral DVT** with severe symptoms, low bleeding risk, or **phlegmasia**.
- **Surgical pulmonary embolectomy:** high-risk PE with contraindication/failure of thrombolysis • VA-ECMO for refractory shock/arrest.
- **IVC filter: only** for **acute proximal DVT or PE + absolute contraindication to anticoagulation** (active bleeding, recent surgery) • **not routinely in patients who can be anticoagulated** (↓ PE early but ↑ DVT long term, no mortality benefit) • use **retrievable** filters and remove when anticoagulation can be started • resume anticoagulation as soon as bleeding risk resolves.
- **Compression stockings: not routinely** recommended to prevent post-thrombotic syndrome (SOX trial) • may relieve symptoms of swelling.

9. Heparin-Induced Thrombocytopenia (HIT)

- **Immune-mediated** (IgG antibodies against **heparin–platelet factor 4 complexes**) → platelet activation → **thrombocytopenia + paradoxical thrombosis** (venous > arterial; limb gangrene, skin necrosis).
- **Timing:** typically **5–10 days** after starting heparin • **rapid onset (within 24 h)** if heparin exposure in the previous **100 days** • delayed-onset after stopping.
- **Platelet fall > 50%** from baseline (nadir usually 20,000–100,000 — rarely severe bleeding).
- More common with **UFH** > LMWH, surgical (cardiac/orthopedic) > medical patients, women.
- **4T score**: **T**hrombocytopenia (degree), **T**iming, **T**hrombosis, o**T**her causes • **low (0–3) → HIT unlikely**, no testing • intermediate/high → **PF4 immunoassay (ELISA)** → confirm with **serotonin release assay** (gold standard functional test).

HIT management: (1) **stop ALL heparin** immediately — UFH, **LMWH** (cross-reactive), **heparin flushes**, heparin-coated catheters • (2) **start a non-heparin anticoagulant** even without thrombosis (high thrombosis risk): **argatroban** (hepatic — renal failure), **bivalirudin** (renal/enzymatic — cardiac surgery/PCI), **fondaparinux**, or a **DOAC** (stable patients without life-threatening thrombosis) • (3) **do NOT give platelet transfusions** (fuel thrombosis) unless bleeding • (4) **do NOT start warfarin until platelets recover (≥ 150,000)** — warfarin alone → venous limb gangrene; if already on warfarin → reverse with vitamin K • start warfarin at low dose (≤ 5 mg) and **overlap ≥ 5 days** until INR therapeutic • (5) **duration:** HIT without thrombosis ~**4 weeks** (to 3 months) • **HIT with thrombosis (HITT): 3 months** • (6) document heparin allergy; avoid heparin for future (except carefully for cardiac surgery when antibodies negative).

10. VTE Prophylaxis

Hospitalized medical patients

- **Assess VTE and bleeding risk** on admission: **Padua Prediction Score ≥ 4** (or IMPROVE ≥ 2) = high risk • IMPROVE-Bleed for bleeding risk.
- **Padua items:** active cancer (3), previous VTE (3), reduced mobility (3), known thrombophilia (3), recent trauma/surgery ≤ 1 month (2), age ≥ 70 (1), HF/respiratory failure (1), acute MI/stroke (1), acute infection/rheumatologic disorder (1), obesity BMI ≥ 30 (1), hormonal therapy (1).
- **High VTE risk + acceptable bleeding risk → pharmacologic prophylaxis**: **enoxaparin 40 mg SC daily** (preferred), **UFH 5,000 units SC q8–12h**, fondaparinux 2.5 mg daily, (dalteparin 5,000 units daily).
- **High bleeding risk** (active bleeding, platelets < 50,000, recent ICH) → **mechanical prophylaxis** (**intermittent pneumatic compression**) → switch to pharmacologic when bleeding risk falls.
- **Low risk → no prophylaxis**, early ambulation • **do not extend prophylaxis after discharge routinely** (selected high-risk: rivaroxaban 10 mg × 31–39 days).
- **Aspirin is not adequate** for medical inpatients.
- **Critically ill:** LMWH or UFH (LMWH preferred) • mechanical if bleeding.
- **Renal impairment (CrCl < 30):** UFH or enoxaparin 30 mg daily.

Surgical patients

- **Caprini score** estimates risk: very low (0) → early ambulation • low (1–2) → mechanical (IPC) • moderate (3–4) → LMWH/UFH or IPC • high (≥ 5) → **LMWH/UFH + IPC** • **extended 4 weeks** after major abdominal/pelvic **cancer surgery**.
- **Hip and knee arthroplasty:** options — **LMWH** (enoxaparin 30 mg q12h starting 12–24 h post-op, or 40 mg daily starting 12 h pre-op), **apixaban 2.5 mg BID**, **rivaroxaban 10 mg daily**, dabigatran (220 mg daily, hip), fondaparinux 2.5 mg, **aspirin** (81 mg BID or 81–325 mg — acceptable especially after initial LMWH), warfarin (INR 2–3), IPC • **duration: minimum 10–14 days, up to 35 days** (especially hip) • start DOACs 12–24 h after surgery.
- **Hip fracture surgery:** LMWH, fondaparinux, UFH, warfarin, aspirin • up to 35 days.
- **Neuraxial anesthesia:** follow timing rules (boxed warning — spinal/epidural hematoma).

Other populations

- **Ambulatory cancer patients** starting chemotherapy with **Khorana score ≥ 2 → apixaban 2.5 mg BID** or **rivaroxaban 10 mg daily** (or LMWH) • multiple myeloma on **thalidomide/lenalidomide** → aspirin (low risk) or LMWH (high risk).
- **Pregnancy:** prior VTE or high-risk thrombophilia → **prophylactic LMWH** antepartum and/or **6 weeks postpartum** • APS with pregnancy loss → LMWH + low-dose aspirin.
- **Long-distance travel** (> 4 h): ambulation, calf exercises, aisle seat, hydration • high-risk → compression stockings or LMWH/DOAC • not aspirin routinely.

11. Special Populations and Complications

- **Pregnancy:** VTE risk ↑ 4–5× (highest **postpartum**) • treat with **LMWH** (weight-based; anti-Xa in some) throughout pregnancy + **≥ 6 weeks postpartum** • switch to UFH ~36 weeks or stop LMWH 24 h before planned delivery/neuraxial anesthesia • **warfarin and DOACs avoided in pregnancy** • **postpartum/breastfeeding: LMWH or warfarin** (DOACs avoided in breastfeeding).
- **Cancer:** oral Xa inhibitors or LMWH • ≥ 6 months and while cancer active • drug interactions with chemotherapy (CYP3A4/P-gp) • thrombocytopenia → dose reduction/hold (platelets < 50,000).
- **Elderly:** ↑ bleeding risk, check renal function, fall risk is not a reason to withhold anticoagulation • avoid NSAIDs/antiplatelets combos.
- **Pediatrics:** LMWH, UFH, rivaroxaban and dabigatran (approved pediatric formulations).
- **Post-thrombotic syndrome** (after DVT): chronic leg pain, edema, venous ulcers → compression for symptoms, exercise, elevation.
- **Chronic thromboembolic pulmonary hypertension (CTEPH)** (~2–4% after PE): persistent dyspnea at **3–6 months** → **V/Q scan** screening • treatment: **pulmonary endarterectomy** (curative), balloon pulmonary angioplasty, **riociguat** (sGC stimulator; CI with PDE5 inhibitors/nitrates; teratogenic REMS), lifelong anticoagulation.
- **Recurrent VTE on anticoagulation:** check adherence, drug interactions, cancer, APS, HIT → switch to **LMWH** (or ↑ LMWH dose by 25% if already on it).

Patient counseling (anticoagulants)

- Take at the **same time daily** • don't stop without talking to provider • missed-dose rules • carry anticoagulant card.
- **Signs of bleeding:** black/tarry stools, blood in urine, vomiting blood/coffee grounds, unusual bruising, nosebleeds/gum bleeding, severe headache (ICH) → seek care.
- **Avoid OTC NSAIDs and aspirin** (use acetaminophen), herbal products (ginkgo, garlic, St John's wort, fish oil high dose).
- Inform all providers/dentists before procedures • soft toothbrush, electric razor • avoid contact sports.
- **Warfarin:** consistent vitamin K diet, regular INR checks, limit alcohol • **rivaroxaban 15/20 mg with food** • **dabigatran in original container, don't open capsules**.

12. VTE Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Apixaban	**10 mg BID × 7 d → 5 mg BID** • extended **2.5 mg BID** • prophylaxis (ortho) 2.5 mg BID	Bleeding (least GI)	No lead-in • used in ESRD • **strong 3A4 + P-gp inhibitors → ↓ 50%** • avoid inducers • cancer VTE ✓
Rivaroxaban	**15 mg BID × 21 d → 20 mg daily with food** • extended **10 mg daily** • prophylaxis 10 mg	Bleeding (GI)	No lead-in • **with food** • avoid CrCl < 15 • avoid strong 3A4+P-gp inhibitors and inducers
Edoxaban	Parenteral 5–10 d → **60 mg daily** (30 mg: CrCl 15–50, ≤ 60 kg, P-gp inhibitors)	Bleeding (GI)	AF: ↓ efficacy CrCl > 95 • cancer VTE ✓
Dabigatran	Parenteral 5–10 d → **150 mg BID**	**Dyspepsia**, GI bleeding	**Avoid CrCl < 30** • don't open capsules • original bottle 4 months • P-gp • **idarucizumab** reversal
Warfarin	**5 mg daily** start (2.5 elderly/liver/HF) • INR 2–3 • overlap ≥ 5 d + INR ≥ 2 × 24 h	**Bleeding**, **skin necrosis**, purple toe, **teratogenic**	**CYP2C9**: amiodarone, fluconazole, metronidazole, SMX-TMP ↑ INR • rifampin, carbamazepine ↓ • vitamin K consistency • APS ✓

Drug	Dose	Key ADRs	Key notes / interactions
UFH	**80 units/kg bolus → 18 units/kg/h** • prophylaxis 5,000 units SC q8–12h	Bleeding, **HIT**, osteoporosis, **hyperkalemia**	aPTT 1.5–2.5× or anti-Xa 0.3–0.7 • unstable PE, thrombolysis, CrCl < 30 • **protamine 1 mg/100 units**
Enoxaparin	**1 mg/kg q12h** or 1.5 mg/kg daily • **CrCl < 30: 1 mg/kg daily** • prophylaxis **40 mg daily** (30 mg if CrCl < 30)	Bleeding, bruising, HIT (less), **spinal hematoma (boxed)**	Anti-Xa in pregnancy/obesity/renal • don't expel air bubble • protamine ~60% • **pregnancy** ✓
Dalteparin	200 units/kg daily • cancer 200 → 150 units/kg after 1 month • prophylaxis 5,000 units	Same	Cancer-associated VTE
Fondaparinux	**< 50 kg 5 mg • 50–100 kg 7.5 mg • > 100 kg 10 mg** daily • prophylaxis 2.5 mg	Bleeding, spinal hematoma (boxed)	**CI CrCl < 30** • no HIT cross-reactivity • no reversal • SVT 2.5 mg × 45 d
Argatroban	**2 mcg/kg/min** (0.5–1.2 hepatic/critically ill)	Bleeding	**HIT** • hepatic → OK in renal failure • **falsely ↑ INR**
Bivalirudin	PCI bolus + infusion • HIT 0.15 mg/kg/h	Bleeding	Renal adjust • HIT with PCI/cardiac surgery
Alteplase	**100 mg IV over 2 h** • arrest 50 mg bolus	**Bleeding, ICH**, angioedema	High-risk PE • hold UFH during infusion, restart when aPTT < 2× ULN
Protamine	1 mg/100 units UFH (max 50 mg) • 1 mg/1 mg enoxaparin	**Hypotension, anaphylaxis**, bradycardia	Slow IV ≤ 5 mg/min • fish allergy, NPH insulin
Vitamin K (phytonadione)	INR > 10: 2.5–5 mg PO • major bleed: 5–10 mg IV slow	**Anaphylaxis (IV)**	Avoid SC/IM • PO preferred unless bleeding
4F-PCC (Kcentra)	25–50 units/kg (by INR) • DOAC bleed 50 units/kg	**Thrombosis** (boxed)	Warfarin major bleeding (+ vitamin K)
Idarucizumab	**5 g IV**	Thrombosis, headache	**Dabigatran** reversal
Andexanet alfa	Low/high dose bolus + infusion (by agent, dose, timing)	**Thromboembolism, MI, stroke** (boxed)	Apixaban/rivaroxaban reversal (limited availability)
Aspirin	81–100 mg daily (extended after VTE) • 81 mg BID ortho prophylaxis	GI bleeding	Less effective than anticoagulants
Riociguat	1 mg TID → 2.5 mg TID	Hypotension, headache, bleeding	CTEPH • **CI with PDE5i/nitrates** • teratogenic REMS

Exam Traps and Frequently Tested Points

1. **Virchow's triad:** stasis, endothelial injury, hypercoagulability • **factor V Leiden** = most common inherited thrombophilia.
2. **Drugs ↑ VTE:** estrogens (COCs, HRT), **tamoxifen, raloxifene**, testosterone, **ESAs**, **thalidomide/lenalidomide**, tofacitinib, L-asparaginase.
3. **Wells DVT ≤ 1 → D-dimer**; ≥ 2 → compression ultrasound • **Wells PE ≤ 4 → D-dimer**; > 4 → **CTPA**.
4. **D-dimer rules OUT** (high sensitivity, low specificity) • age-adjusted cut-off = **age × 10** (> 50 years) • not useful in high probability.
5. **PERC rule** (8 items, low probability only) → no testing if all negative.
6. **CTPA** = test of choice for PE • **V/Q scan** if CKD, contrast allergy, pregnancy • unstable → **bedside echo** (RV dysfunction).
7. **ECG: sinus tachycardia most common**; S1Q3T3 classic • ABG: hypoxemia + respiratory alkalosis • troponin/BNP for risk stratification.
8. **High-risk PE (shock/hypotension) → thrombolysis (alteplase 100 mg over 2 h)** • intermediate-high → anticoagulate + monitor (rescue lysis) • low risk (**sPESI 0**) → outpatient DOAC.

9. **Hold UFH during alteplase**, restart without bolus when aPTT < 2× ULN • know **absolute contraindications** (prior ICH, stroke < 3 months, active bleeding, dissection).
10. **DOACs preferred over warfarin** for most VTE • **apixaban/rivaroxaban need no lead-in** • **dabigatran/edoxaban need 5–10 days parenteral first**.
11. **Apixaban 10 mg BID × 7 days → 5 mg BID** (extended 2.5 mg BID) • **rivaroxaban 15 mg BID × 21 days → 20 mg daily with food** (extended 10 mg).
12. **Dabigatran 150 mg BID**: avoid CrCl < 30, dyspepsia, don't open capsules, original bottle • **edoxaban 60 mg** (30 mg if CrCl 15–50, ≤ 60 kg, P-gp inhibitors).
13. **Cancer VTE → apixaban/rivaroxaban/edoxaban** (or LMWH, especially GI/GU cancers) • **APS → warfarin** (no DOACs) • **pregnancy → LMWH** • **CrCl < 30 → UFH/warfarin** (apixaban possible).
14. **Duration:** provoked → **3 months** • unprovoked → 3 months then **extended** if bleeding risk low • cancer → while active • 2nd unprovoked → indefinite.
15. **Isolated distal DVT** → serial ultrasound 2 weeks if no severe symptoms • **subsegmental PE** without DVT → surveillance • **SVT ≥ 5 cm → fondaparinux 2.5 mg × 45 days**.
16. **Warfarin:** overlap with parenteral **≥ 5 days AND INR ≥ 2 for 24 h** • start 5 mg (2.5 elderly) • goal INR 2–3 • factors II, VII, IX, X, proteins C and S.
17. **Warfarin skin necrosis** (protein C deficiency, days 3–8) • **teratogenic** • consistent vitamin K diet.
18. **Warfarin ↑ INR:** amiodarone, fluconazole, metronidazole, SMX-TMP, fluoroquinolones • **↓ INR:** rifampin, carbamazepine, phenytoin, St John's wort, vitamin K.
19. **INR 4.5–10 no bleeding → hold, no vitamin K • INR > 10 → oral vitamin K 2.5–5 mg • major bleeding → 4F-PCC + vitamin K 5–10 mg IV**.
20. **Reversal:** dabigatran → **idarucizumab 5 g** • Xa inhibitors → andexanet alfa or **4F-PCC** • **UFH → protamine 1 mg/100 units** • LMWH partial (~60%) • fondaparinux none.
21. **UFH:** 80 units/kg → 18 units/kg/h • aPTT 1.5–2.5× • ADRs: HIT, osteoporosis, **hyperkalemia** • preferred in unstable PE, renal failure, possible procedures.
22. **Enoxaparin: 1 mg/kg q12h** (1 mg/kg daily if CrCl < 30) • prophylaxis 40 mg daily (30 mg if CrCl < 30) • anti-Xa in pregnancy/obesity/renal • **boxed spinal hematoma**.
23. **Fondaparinux:** weight-based 5/7.5/10 mg • **CI CrCl < 30** • no HIT cross-reactivity.
24. **DOAC DDIs:** avoid **strong 3A4 + P-gp inhibitors** (ketoconazole, ritonavir) and **inducers** (rifampin, carbamazepine, phenytoin, St John's wort) • NSAIDs/SSRIs/antiplatelets ↑ bleeding.
25. **DOACs contraindicated** with mechanical heart valves and moderate–severe mitral stenosis; avoid in APS, pregnancy, breastfeeding.
26. **HIT:** platelets ↓ > 50% at **5–10 days** (or < 24 h if exposure within 100 days) + thrombosis • **4T score** → PF4 ELISA → serotonin release assay.
27. **HIT treatment: stop all heparin (including LMWH and flushes)** → **argatroban** (renal failure) / bivalirudin / fondaparinux / DOAC • **no platelet transfusions** • **no warfarin until platelets ≥ 150,000** (overlap ≥ 5 days) • argatroban falsely ↑ INR.
28. **IVC filter** only if acute VTE + **absolute contraindication to anticoagulation** • retrievable • not routine with anticoagulation.
29. **Compression stockings not routinely** to prevent post-thrombotic syndrome • early ambulation encouraged • home treatment for DVT/low-risk PE.
30. **Medical inpatient prophylaxis: Padua ≥ 4 → enoxaparin 40 mg daily** or UFH 5,000 q8–12h • bleeding risk → IPC • **aspirin not adequate** • no routine extended prophylaxis.
31. **Ortho (hip/knee):** LMWH, apixaban 2.5 BID, rivaroxaban 10 daily, aspirin, fondaparinux, warfarin • **10–14 days up to 35 days**.
32. **Cancer outpatients Khorana ≥ 2** → apixaban 2.5 mg BID or rivaroxaban 10 mg • **myeloma on lenalidomide** → aspirin or LMWH.
33. **Pregnancy:** LMWH throughout + **≥ 6 weeks postpartum** (≥ 3 months total) • warfarin OK postpartum/breastfeeding • DOACs avoid.
34. **CTEPH** → persistent dyspnea 3–6 months after PE → V/Q scan • pulmonary endarterectomy, **riociguat** (CI with PDE5i/nitrates).
35. **Counseling:** signs of bleeding, avoid NSAIDs/aspirin, herbal products, rivaroxaban with food, dabigatran original bottle, never double doses.

> **Note:** doses and recommendations follow CHEST (2016/2021), ASH (2018–2020), ESC 2019 PE, and ISTH guidance. If your course slides or reference use slightly different numbers, **use your course numbers for the exam**.